MODEL EVALUATION IN MACHINE LEARNING

What is model evaluation

Model evaluation is used to **measure the performance of a machine learning model**

It helps determine **how accurate the model's predictions are**

It compares **predicted results with actual results**

Train Test Split

DATASET
(e.g., 1000 Records)

TRAINING DATA (80%)
- 800 Records

used to **train** the model

TESTING DATA (20%)
- 200 Records

used to **evaluate**
used to evaluate the model. Helps check how well the model performs on new data.

Confusion Matrix

- A **Confusion Matrix** is used to evaluate classification models.
- It shows how many predictions are correct and incorrect.

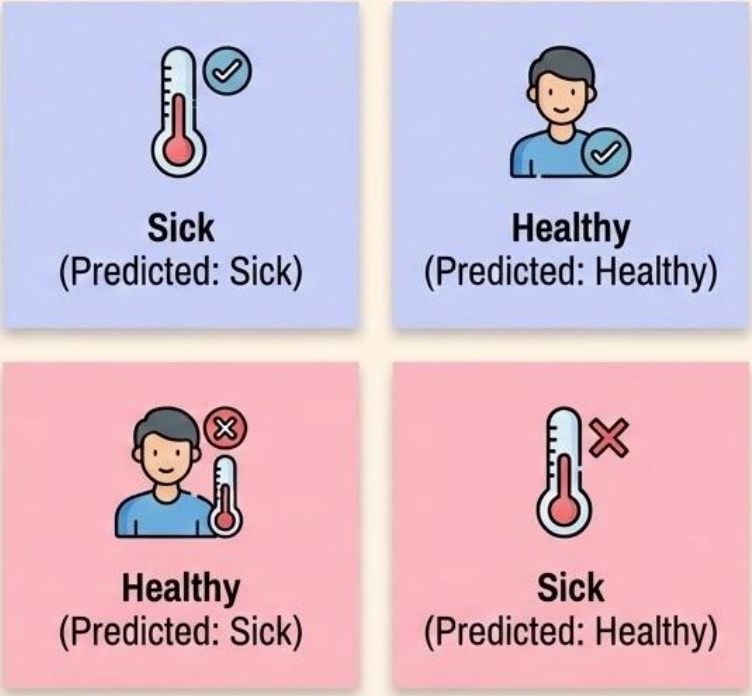
Confusion Matrix Terminology

True Positive (TP) ✓
Model correctly predicts positive.
Example: Sick person predicted as sick.

True Negative (TN) ✓
Model correctly predicts negative.
Example: Healthy person predicted healthy.

False Positive (FP) ✗
Model predicts positive but actually negative.
Example: Healthy person predicted sick.

False Negative (FN) ✗
Model predicts negative but actually positive.
Example: Sick person predicted healthy.



Accuracy

Accuracy measures **how many predictions are correct**

ACCURACY MEASURES CORRECTNESS

How many predictions are correct

$$\textbf{Accuracy} = \frac{\text{Correct Predictions}}{\text{Total Predictions}}$$

Example: 100 Total Predictions

90 correct results, Accuracy = **90%**

Precision

Precision measures **how many predicted positives are actually correct**
It focuses on the **quality of positive predictions**.

Formula: Precision = TP / (TP + FP)

Example Calculation: Precision = 8 / 10 = **0.8 (80%)**

Recall

Recall measures how many **actual positives** were **correctly identified**
It focuses on **detecting all positive cases**

Formula: Recall = TP / (TP + FN)

Example: If there are 20 spam emails and the model detects 15 of them,
Recall = 15 / 20 = **0.75** (75%)

Specificity

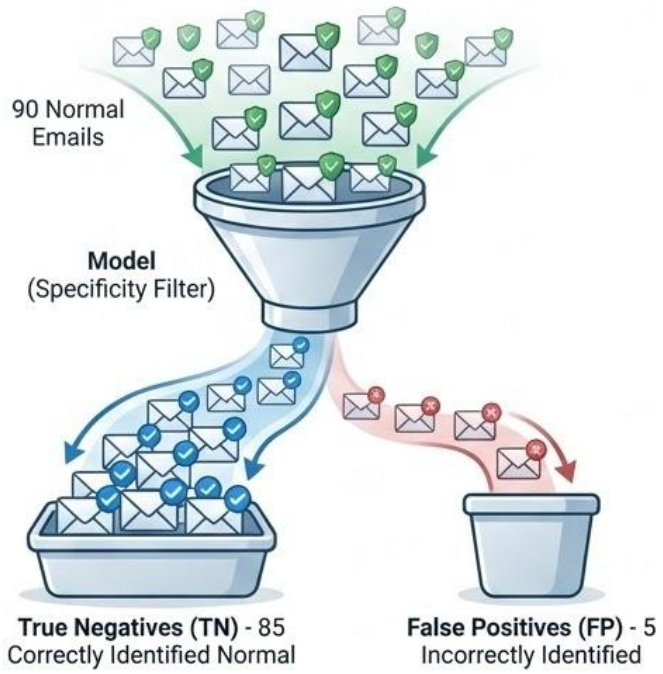
- Specificity measures how well the model detects **negativly identified negative cases**.
- It focuses on detecting all positive cases

Formula:

Specificity = TN / (TN + FP)

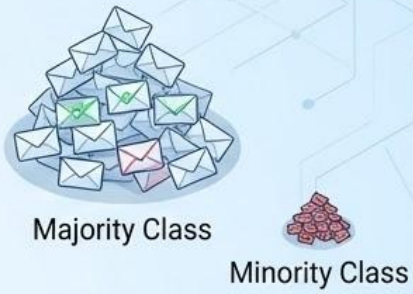
Example: Out of **90 normal emails**, the model **correctly identifies 85**.

Specificity = 85 / 90



F1 Score

- F1 Score is the **balance between Precision and Recall**
- It is useful when the dataset is **imbalanced**



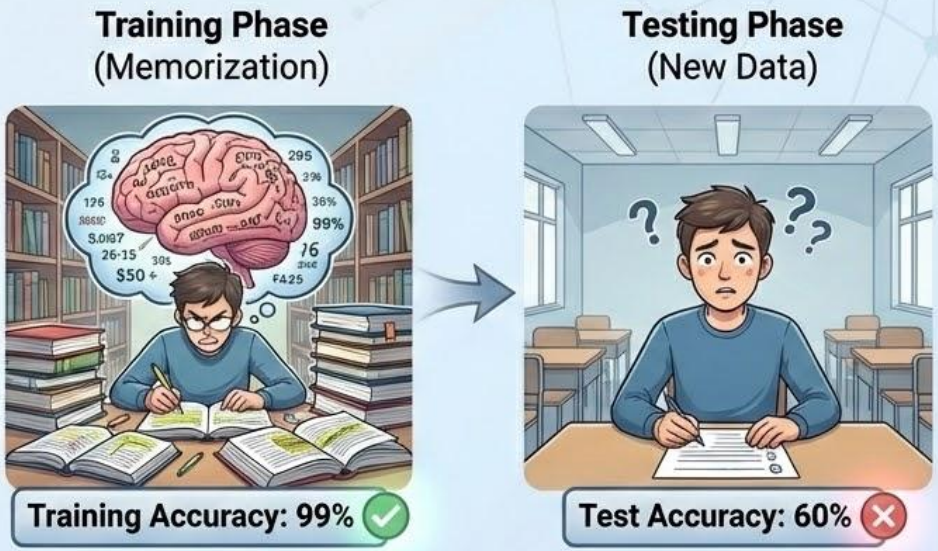
Imbalanced Dataset

Formula: **F1 Score = 2 × (Precision × Recall) / (Precision + Recall)**

		Positive	Negative	
Actual Class	**Positive**	True Positive (TP)	False Negative (FN) **Type II Error**	**Sensitivity** $\frac{TP}{(TP+FN)}$
	Negative	False Positive (FP) **Type I Error**	True Negative (TN)	**Specificity** $\frac{TN}{(TN+FP)}$
		Precision $\frac{TP}{(TP+FP)}$	**Negative Predictive Value** $\frac{TN}{(TN+FN)}$	**Accuracy** $\frac{TP+TN}{(TP+TN+FP+FN)}$

Overfitting

Overfitting occurs when a model learns training data **too well**, capturing noise instead of the underlying pattern.

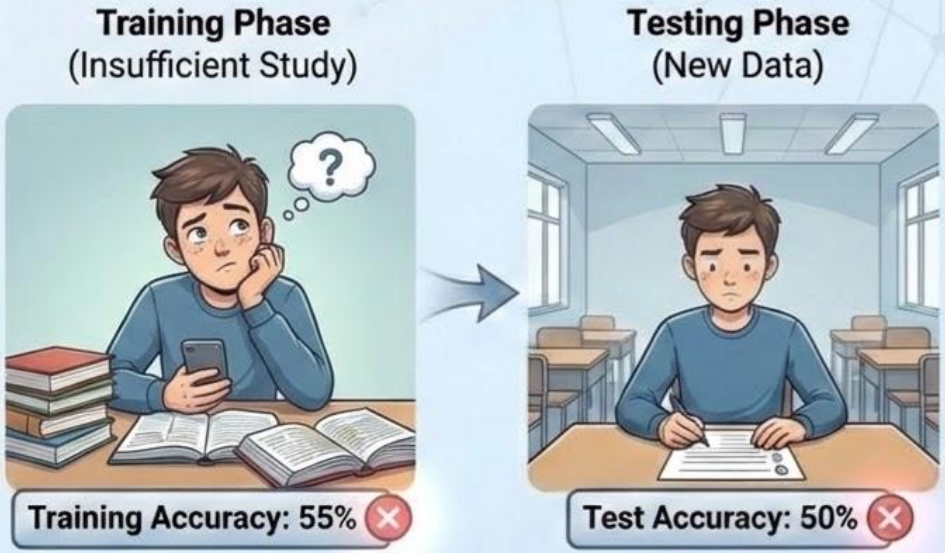


Characteristics:

- Very high **training accuracy**.
- Poor performance on **new, unseen data**.

Underfitting

Underfitting occurs when the model **fails to learn patterns**.



Characteristics:

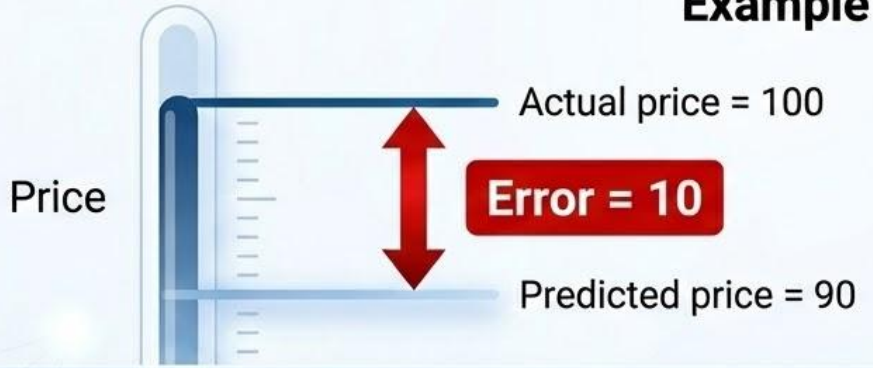
- Low **training accuracy**.
- Low **testing accuracy**.

Mean Absolute Error (MAE)

Definition:
MAE measures the **average error in predictions.**

$$MAE = \text{average}(|\textbf{actual} - \textbf{predicted}|)$$

Example



MAE calculates the **average of all such errors.**

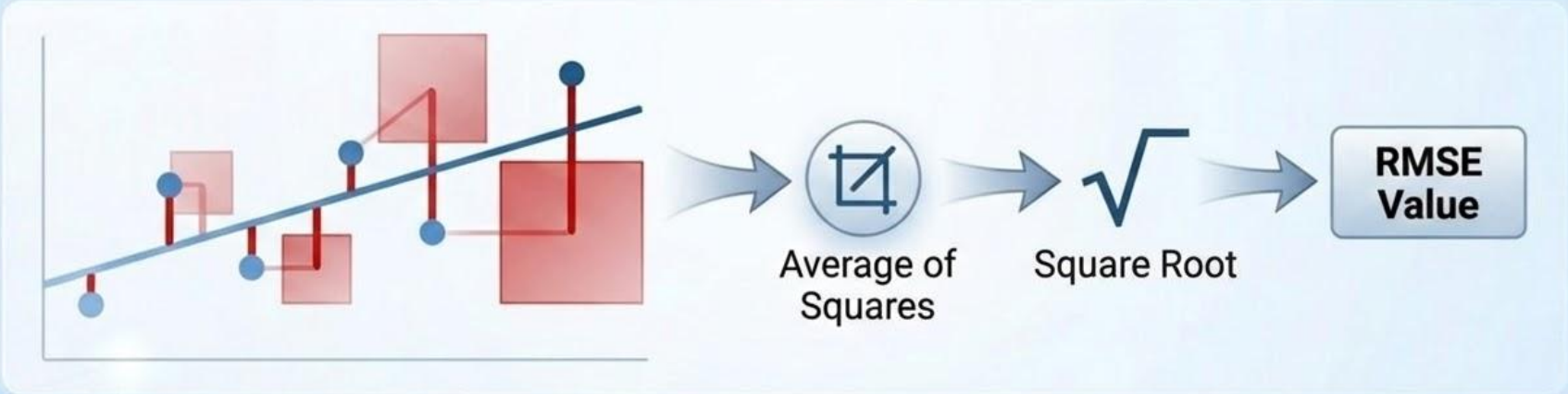
Lower MAE = better model.

Root Mean Squared Error (RMSE)

Definition:
RMSE measures the **square root** of average squared errors.

Formula:

$$RMSE = \sqrt{(\text{average}((\textbf{actual} - \textbf{predicted})^2)}$$



Lower RMSE = better model.

R^2 Score

R^2 score measures how **well the model explains the data**.

Formula

RMSE = √(average((actual – predicted)²))

Important point

RMSE penalizes **large errors more than MAE**.

Lower RMSE = better model.

Example

$R^2 = 0.85$
Means the model explains 85% of the variation in data.

Higher R^2 = better model.